Tuberculosis (TB) — Comprehensive Guide

1. Overview and Global Burden

Tuberculosis (TB) is a bacterial infection caused by Mycobacterium tuberculosis (MTB), primarily affecting the lungs but capable of affecting virtually every organ system. TB remains the world's deadliest infectious disease from a single pathogen, responsible for approximately 1.6 million deaths annually.

India carries the highest TB burden globally — 28% of the world's TB cases. In 2022, India notified approximately 2.4 million new TB cases. Despite significant progress, TB elimination (incidence <1 case per million population) remains a distant goal.

India has set an ambitious target to eliminate TB by 2025 under the National Strategic Plan for TB Elimination, five years ahead of the WHO End TB Strategy's 2030 deadline. The Pradhan Mantri TB Mukt Bharat Abhiyaan (2022) involves community participation in supporting TB patients.

2. Microbiology and Transmission

Mycobacterium tuberculosis is an aerobic, slow-growing, acid-fast bacillus (AFB). Its thick waxy cell wall (mycolic acid) makes it resistant to many antibiotics and the immune system. It can survive in dried sputum for weeks.

Transmission: TB spreads primarily through the air when people with active pulmonary TB cough, sneeze, sing, or speak, releasing infectious aerosol droplet nuclei (1-5 micrometers) that can remain suspended for hours. A single cough can generate 3,000 infectious particles.

Risk of infection depends on: proximity and duration of exposure, infectiousness of the source case (sputum smear-positive cases are most infectious), ventilation of the shared space, and susceptibility of the exposed individual (HIV, malnutrition, diabetes, immunosuppression markedly increase risk).

3. Latent vs Active TB

After inhaling MTB, the immune system usually contains the bacteria: granulomas form and the bacteria enter a latent state. Latent TB infection (LTBI): bacteria dormant, person not sick, not infectious, tuberculin skin test (TST) or IGRA positive.

Lifetime risk of LTBI reactivating to active TB is approximately 10% (5% in first 2 years, 5% over lifetime) in immunocompetent individuals. In HIV-positive individuals, the annual risk is 5-10%.

Active TB: bacteria multiply, person is sick and (if pulmonary) infectious. Factors promoting reactivation: HIV infection (most important), malnutrition, diabetes (2-3x risk), silicosis, end-stage renal disease, immunosuppressive therapy, alcohol abuse, smoking, and extreme age.

4. Symptoms and Extrapulmonary TB

Pulmonary TB (most common form, 60-70%): persistent productive cough (>2 weeks), haemoptysis, chest pain, fever (often low-grade, evening rise), drenching night sweats, unexplained weight loss, fatigue, and anorexia. Advanced disease: breathlessness, formation of cavities.

Extrapulmonary TB (30-40%): Lymph node TB (most common extrapulmonary form in India) — painless swelling of cervical nodes. Pleural TB — pleural effusion with exudative fluid. Bone and joint TB — Pott's disease (spinal TB causing gibbus deformity, cord compression).

TB Meningitis (TBM) — severe headache, neck stiffness, photophobia, confusion, cranial nerve palsies, hydrocephalus. Very high mortality and morbidity. CSF: lymphocytic pleocytosis, high protein, low glucose, AFB smear/culture/Xpert.

Abdominal TB — peritonitis (ascites with insidious onset), intestinal TB (ileocaecal region, causing obstruction, malabsorption), hepatic and splenic involvement. Miliary TB — haematogenous dissemination to all organs; minute bilateral granulomas on chest X-ray (millet seed pattern).

5. Diagnosis

Sputum smear microscopy (AFB smear): inexpensive, rapid, widely available. Detects MTB in sputum. Sensitivity 45-80% (lower in HIV). Requires 2 sputum samples (spot and early morning).

Xpert MTB/RIF assay (GeneXpert): molecular test (PCR-based). 90-minute turnaround. Sensitivity ~89% vs culture as reference standard, near-100% specificity. Simultaneously detects rifampicin resistance. Now available in most district TB centres under NTEP.

Sputum culture (MGIT Liquid culture or LJ solid medium): gold standard. Higher sensitivity. MGIT culture takes 1-6 weeks. Essential for DST (drug susceptibility testing) in resistant TB.

Chest X-ray: characteristic findings — upper lobe infiltrates, cavities, hilar lymphadenopathy, miliary pattern. Not diagnostic alone. Normal CXR does not exclude extrapulmonary TB.

IGRA (Interferon-Gamma Release Assay, e.g. QuantiFERON-TB Gold): blood test for LTBI. More specific than TST (not affected by BCG vaccination). Not diagnostic of active TB.

6. Treatment Regimens

	Drugs (Abbreviation)	Frequency
	Isoniazid + Rifampicin + Pyrazinamide + Ethambutol (HRZE)	Daily
	Isoniazid + Rifampicin (HR)	Daily
	Isoniazid + Rifampicin (HR)	Daily
	Bedaquiline + Pretomanid + Linezolid ± Moxifloxacin	Daily
	Isoniazid + Rifapentine (weekly)	Weekly

All TB treatment in India is provided free under the NTEP. Fixed-dose combinations (FDCs) are used to simplify regimens and prevent monotherapy (reducing resistance). Pyridoxine (B6) 25-50 mg/day is given with isoniazid to prevent peripheral neuropathy.

Monitoring during treatment: monthly sputum smear/culture during intensive phase. LFTs at baseline and if symptoms develop (isoniazid, rifampicin, and pyrazinamide are all hepatotoxic). Visual acuity testing (ethambutol causes optic neuritis).

7. Drug-Resistant TB

MDR-TB (Multi-drug resistant TB): resistant to at least isoniazid AND rifampicin — the two most potent first-line drugs. In India, approximately 6% of new TB cases and 11-13% of previously treated cases are MDR-TB.

XDR-TB (Extensively drug-resistant TB): older definition — MDR + fluoroquinolone + injectable resistance. Newer WHO 2021 definition — MDR/RR-TB + resistance to any fluoroquinolone. Pre-XDR-TB is now defined.

Treatment: BPaL regimen — Bedaquiline + Pretomanid + Linezolid ± Moxifloxacin for 6 months (TB-PRACTECAL, ZeNix, NExT trials). This represents a revolution in MDR-TB treatment, replacing 18-24 month regimens with injections with shorter all-oral regimens with fewer side effects.

Key side effects to monitor: bedaquiline — QT prolongation (ECG monitoring monthly), hepatotoxicity; linezolid — peripheral neuropathy, optic neuritis, myelosuppression (CBC monthly); delamanid — QT prolongation.

8. TB and HIV Co-infection

HIV is the single most powerful risk factor for TB. People living with HIV (PLHIV) are 18x more likely to develop TB than HIV-negative individuals. TB is the leading cause of death in PLHIV.

Diagnosis of TB in PLHIV is challenging — atypical presentations, lower rates of cavitation, sputum smear often negative, and extrapulmonary TB much more common. Urine LAM (lipoarabinomannan) test is useful for bedside rapid diagnosis in PLHIV with low CD4.

Treatment: treat TB first, start ART 2-8 weeks later (except TB meningitis — defer ART 4-6 weeks). Rifampicin is a potent inducer of cytochrome P450 and significantly reduces levels of many antiretroviral drugs — dose adjustment of efavirenz is required; rifabutin is an alternative in some regimens.

Immune Reconstitution Inflammatory Syndrome (IRIS): paradoxical worsening of TB symptoms after ART initiation due to restored immune response. Treat with NSAIDs or corticosteroids. Usually self-limiting.

9. TB in Special Populations and Infection Control

TB in children: presents more commonly as extrapulmonary or disseminated TB. Bacteriological confirmation harder (paucibacillary disease). Clinical algorithm (symptoms + contact history + X-ray + TST/IGRA) used. Xpert Ultra improves yield. BCG vaccination at birth offers 70-80% protection against severe forms (miliary, meningitis) in children.

TB in pregnancy: rifampicin and isoniazid are safe throughout pregnancy. Pyrazinamide — use if benefits outweigh risks. Streptomycin absolutely contraindicated (causes fetal ototoxicity). TB in pregnancy associated with higher maternal and neonatal mortality. Treat without delay.

Infection control: administrative controls (triage, fast-tracking, separation of infectious patients, cough hygiene). Environmental controls (natural and mechanical ventilation — 12+ air changes per hour in TB wards; UV germicidal irradiation). Personal protective equipment (N95 respirators for HCWs — NOT surgical masks).

Healthcare worker TB: HCWs have 2-5x higher TB risk than the general population. Annual IGRA/TST screening, LTBI treatment for positive results in high-risk HCWs.

10. Nikshay, NTEP, and the Path to Elimination

Nikshay is India's web-based TB patient management system under NTEP. All TB patients must be notified on Nikshay. The Nikshay Poshan Yojana provides Rs 500/month nutritional support to TB patients for the duration of treatment.

The NTEP uses Directly Observed Treatment Short-course (DOTS) to ensure treatment adherence. Community volunteers (Nikshay Mitras) provide support and monitoring. Video-Observed Therapy (VOT) is being scaled up.

Key challenges: catastrophic costs (TB pushes ~50 million Indians into poverty annually), stigma, nutritional deficiency (TB prevalence is highest in malnourished populations — BMI <18.5 doubles TB risk), access in tribal and hard-to-reach areas, and missing the private sector (60% of TB patients first seek care in the private sector in India).

India's path to TB elimination requires: universal drug susceptibility testing, accelerated scale-up of preventive therapy, addressing social determinants (nutrition, poverty, housing), active case-finding campaigns, and engaging the private sector under the 99DOTS and private provider engagement initiatives.